

and `y` and the value returned by the function `sub` will be 2 in this case is displayed. In the next line the function `sub` is again called by the variables `x` and `y`. This time the return value is stored in the variable `v` after adding 4 to it. So the value of `v` will be 6 and is displayed by the next line.

5.3. Call By Reference

Before understanding Call by reference let us discuss what happens when arguments are passed by value. Generally, any argument passed to the function, is passed by value. Therefore, whenever a function is called, along with its parameters a copy of the values of the variables is passed, but not the actual variables. Consider the following example.

```
int m=5, y=3, k;  
k= adding (m,y);
```

Here, the function `adding` is called and the values of the respective variables `m` and `y`, 5 and 3 but the original variables are never passed. However, in some situations, the need arises to manipulate the value of a certain variable from within a function. This can be explained with the help of the following program.

```
// passing parameters by reference  
  
#include <iostream.h>  
void duplet(int& m, int& n, int& o)  
{  
    m*=2;  
    n*=2;  
    o*=2;  
}  
int main ()  
{  
    int x=2, y=7, z=9;
```